

MUNICIPAL COMPLAINT MANAGEMENT SYSTEM

Software Requirements / Requirements Report

Project Type: SDLC / Software Engineering Academic Project

1. Introduction

1.1 Project Title

Municipal Complaint Management System (MCMS)

1.2 Purpose

The Municipal Complaint Management System is a web-based application designed to provide an efficient and transparent platform for citizens to report problems related to municipal services and for municipal authorities to manage, assign, monitor, and resolve those complaints. The system supports complaint tracking, notifications, feedback, reporting, and selected AI-assisted features such as automatic complaint classification and priority prediction.

2. Problem Statement

Traditional municipal complaint systems often rely on physical complaint offices, telephone calls, emails, or disconnected online systems. These approaches can make it difficult for citizens to track complaints and for authorities to efficiently manage large numbers of complaints.

- Lack of centralized complaint management.
- Difficulty tracking complaint status.
- Delayed response to complaints.
- Complaints being assigned to the wrong department.
- Lack of accountability for field workers.
- Difficulty identifying overdue complaints.
- Lack of evidence regarding complaint resolution.
- Poor communication between citizens and municipal authorities.
- Difficulty generating performance reports.
- Duplicate complaints regarding the same civic issue.

Therefore, a centralized Municipal Complaint Management System is proposed to improve transparency, accountability, communication, and efficiency.

3. Objectives

- Provide citizens with an easy platform for registering municipal complaints.
- Provide each complaint with a unique complaint ID.
- Allow citizens to track complaint status.
- Categorize complaints according to municipal services.
- Route complaints to appropriate municipal departments.

- Allow municipal officers to verify and manage complaints.
- Allow officers to assign complaints to field workers.
- Allow field workers to update complaint progress.
- Allow workers to upload proof of completed work.
- Allow officers to verify complaint resolution.
- Notify citizens about important status changes.
- Collect citizen feedback after resolution.
- Identify overdue complaints and support escalation.
- Provide dashboards and reports for municipal authorities.
- Use AI to assist with complaint classification and priority prediction.
- Improve transparency and accountability in municipal service delivery.

4. Scope of the System

4.1 Citizen Scope

- Register an account
- Login/logout
- Manage profile
- Submit complaints
- Select complaint categories
- Provide descriptions and location
- Upload photographs/evidence
- Track complaint status
- View complaint history
- Receive notifications
- Request reopening of unresolved issues
- Provide ratings and feedback

4.2 Municipal Officer Scope

- View, search and filter complaints
- Verify complaints
- Reject invalid complaints
- Assign departments and field workers
- Set or modify priority
- Set resolution deadlines
- Monitor progress
- Verify resolution evidence
- Reopen complaints

- Escalate overdue complaints
- Generate reports

4.3 Field Worker Scope

- Login
- View assigned complaints
- View complaint details and location
- Update work progress
- Upload before/after images
- Mark assigned complaints completed
- View pending and completed assignments

4.4 Administrator Scope

- Manage users
- Manage officers
- Manage field workers
- Manage departments
- Manage complaint categories
- Manage system settings
- Monitor system activity
- Generate overall reports

5. Stakeholders

Stakeholder	Role
Citizens	Submit and track complaints
Municipal Officers	Manage and monitor complaints
Field Workers	Resolve assigned complaints
System Administrator	Manage the overall system
Municipal Departments	Receive and resolve complaints
Municipal Management	Monitor performance and reports

6. User Roles

Role	Description
Citizen	Can submit and track their own complaints.
Municipal Officer	Can manage complaints within their assigned municipal department.
Field Worker	Can view and resolve complaints assigned to them.
Administrator	Has system-level management privileges.

7. Functional Requirements

ID	Requirement
FR-01: User Registration	The system shall allow citizens to create an account using name, email, phone number, address and password.
FR-02: User Login	The system shall authenticate registered users and redirect them to the dashboard corresponding to their role.
FR-03: Role-Based Access Control	The system shall provide different permissions for Citizen, Municipal Officer, Field Worker and Administrator.
FR-04: Complaint Registration	Citizens shall submit title, description, category, location, priority where applicable, and supporting evidence. A user shall be able to register multiple complaints.
FR-05: Complaint Categorization	The system shall support categories such as Roads, Garbage, Water Supply, Drainage, Electricity/Streetlights, Sewerage, etc.
FR-06: Location Management	Citizens shall provide the problem location using a manual address, map-based location, or GPS-based location.
FR-07: Evidence Upload	Citizens shall upload photographs or other supported evidence associated with a complaint.
FR-08: Complaint ID Generation	The system shall automatically generate a unique complaint identifier, e.g. MCMS-2026-00125.
FR-09: Complaint Verification	Municipal officers shall review complaints, verify or reject them, provide rejection reasons, and modify category or location if necessary.
FR-10: Department Assignment	Verified complaints shall be assigned to an appropriate municipal department.
FR-11: Worker Assignment	Officers shall assign complaints to field workers with complaint ID, worker, department, priority, deadline and assigned date.
FR-12: Complaint Status Management	The system shall maintain statuses such as Submitted, Under Review, Verified, Assigned, In Progress, Resolved, Closed, etc.
FR-13: Complaint Tracking	Citizens shall track the current status and view a status timeline.
FR-14: Worker Progress Updates	Field workers shall update the progress of assigned complaints.
FR-15: Resolution Evidence	Workers shall upload evidence of completed work, preferably before/after images.
FR-16: Resolution Verification	Officers shall approve resolution, reject resolution, request additional work, or reopen the complaint.
FR-17: Complaint Reopening	Citizens may request reopening when an issue has not been properly resolved; officers review the request.
FR-18: Notifications	The system shall notify users about submission, verification, rejection, assignment, work start, resolution and reopening.
FR-19: Feedback and Rating	Citizens shall provide a 1–5 rating and optional comments after resolution.
FR-20: Search and Filtering	Officers and administrators shall filter complaints by ID, category, status, priority, department, worker, location and assigned date.
FR-21: Deadline Management	Officers shall specify resolution deadlines and the system shall identify complaints exceeding their deadlines.
FR-22: Escalation	The system shall identify overdue complaints and notify responsible officers; repeatedly overdue complaints may be escalated to higher management.
FR-23: Dashboard	The system shall provide role-specific dashboards for citizens, officers, workers and administrators.
FR-24: Reports	The system shall generate reports by category, department, location, status, priority, date, resolution time, worker, etc.

8. AI-Based Functional Requirements

ID	Requirement
AI-FR-01: Automatic Complaint Classification	The system may analyze citizen-entered text and suggest a category/subcategory. Users and officers can modify the suggestion.
AI-FR-02: Priority Prediction	The system may analyze complaint information and suggest Critical, High, Medium or Low priority. The municipality may override the suggestion.
AI-FR-03: Duplicate Complaint Detection	The system may compare new complaints with existing complaints and identify potentially similar complaints. The system may suggest merging similar complaints.

9. Non-Functional Requirements

ID	Requirement
NFR-01: Performance	Normal user requests should respond within approximately 3 seconds under normal load.
NFR-02: Security	Use secure authentication, password hashing, role-based authorization, input validation and protection of sensitive data.
NFR-03: Usability	Provide simple navigation, clear controls, easy complaint submission, clear status indicators and responsive design.
NFR-04: Reliability	Maintain complaint information reliably and prevent accidental data loss during normal operation.
NFR-05: Availability	The system should be available throughout the day except during planned maintenance.
NFR-06: Scalability	The system should support increasing numbers of citizens, complaints, departments, workers and officers.
NFR-07: Maintainability	Use modular architecture so new features, categories and departments can be added with minimal changes.
NFR-08: Compatibility	Support modern browsers such as Chrome, Edge and Firefox.
NFR-09: Data Privacy	Personal and complaint information shall only be accessible to authorized users.
NFR-10: Backup and Recovery	The database should be backed up periodically so important complaint information can be recovered after system failure.

10. External Interface Requirements

10.1 User Interface

- Citizen: registration, login, dashboard, complaint submission, tracking, notifications and feedback.
- Officer: dashboard, complaint management, assignment, reports and resolution verification.
- Worker: dashboard, assigned complaints, details, progress updates and evidence upload.
- Administrator: user, department and category management, dashboard and reports.

10.2 Software Interfaces

- MongoDB database
- Authentication system
- File/image storage service
- Mapping service
- Notification service

10.3 Hardware Interfaces

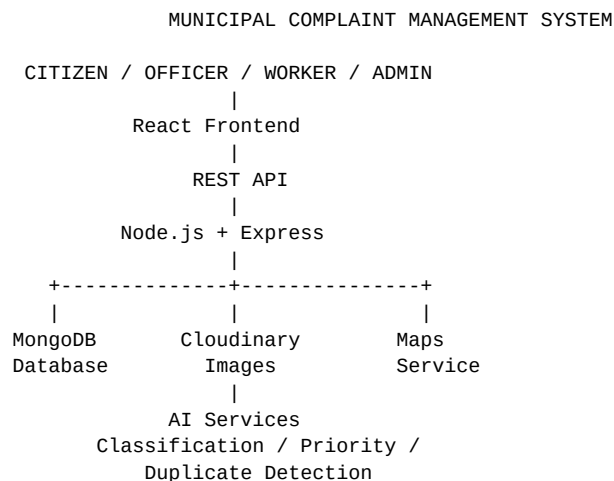
- Desktop computers
- Laptops
- Tablets
- Smartphones
- Smartphone camera for complaint evidence

11. Proposed Technology Stack

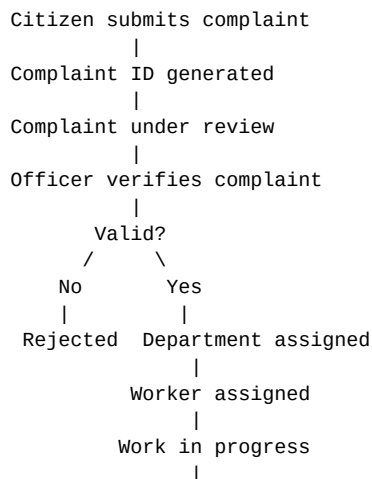
Layer	Technology
Frontend	React.js
Styling	CSS / Tailwind CSS

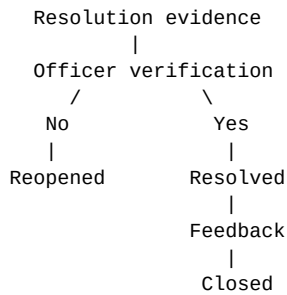
Layer	Technology
Backend	Node.js
Backend Framework	Express.js
Database	MongoDB
Database Management	MongoDB Compass
Authentication	JWT
Password Security	bcrypt
File Upload	Multer
File Storage	Cloudinary
Maps	Leaflet + OpenStreetMap
Charts	Recharts
API Testing	Postman
Version Control	Git + GitHub
Development Environment	VS Code

12. System Architecture



13. Complaint Lifecycle





14. Business Rules

- A citizen must be authenticated before submitting a non-anonymous complaint.
- Every complaint must have a unique complaint ID.
- Every complaint must belong to a valid category.
- Only authorized municipal officers can verify complaints.
- Only authorized officers can assign complaints to workers.
- A field worker can update only complaints assigned to them.
- A worker should provide resolution evidence before marking a complaint completed.
- Only an authorized officer can verify the final resolution.
- Citizens can provide feedback only after the complaint has been resolved.
- Citizens can request reopening after resolution.
- AI-generated classifications and priorities are recommendations; officers retain final authority.
- Users cannot access information or functionality outside their assigned role.

15. Assumptions

- Citizens have access to the internet.
- Users provide reasonably accurate information.
- Municipal departments and workers are registered in the system.
- Municipal officers are responsible for verifying complaints.
- Field workers have access to smartphones or computers.
- Internet connectivity is available during complaint processing.
- Uploaded images comply with supported formats and size limits.
- AI predictions are advisory and may require human verification.

16. Constraints

- Internet connectivity is required.
- AI prediction quality depends on available training data.
- Map and cloud storage services may have usage limits.
- Municipal staff must update complaint status accurately.
- Large numbers of image uploads may increase storage requirements.

- Some advanced features may require third-party APIs.
- The initial project may be a prototype rather than a production municipal deployment.

17. Feasibility Analysis

17.1 Technical Feasibility

The system can be developed using widely available technologies such as React.js, Node.js, Express.js and MongoDB. Therefore, the project is technically feasible.

17.2 Economic Feasibility

The prototype can use open-source technologies and free/developer-tier services, keeping development cost low.

17.3 Operational Feasibility

The system is designed to simplify complaint registration for citizens and complaint management for municipal authorities. Therefore, it is operationally feasible.

17.4 Schedule Feasibility

The project can be developed in phases: requirement analysis, system design, database design, frontend development, backend development, integration, testing and deployment.

18. Security Requirements

- Secure authentication.
- Password hashing.
- JWT-based authorization.
- Role-based access control.
- Input validation.
- File type validation.
- File size restrictions.
- Protection against unauthorized API access.
- Secure handling of user information.
- Proper session/token management.

19. Testing Requirements

Unit Testing

Individual functions and modules shall be tested independently.

Integration Testing

Interactions between frontend/backend, backend/database and external services shall be tested.

System Testing

The complete complaint lifecycle shall be tested from submission through closure.

Security Testing

Unauthorized access and role permissions shall be tested.

Usability Testing

The system shall be evaluated for ease of complaint submission and tracking.

User Acceptance Testing

Representative municipal users shall evaluate whether the system meets specified requirements.

20. Success Criteria

- Citizens can register and log in successfully.
- Citizens can submit complaints.
- Every complaint receives a unique ID.
- Complaints can be categorized and assigned.
- Officers can verify complaints.
- Workers can receive assignments.
- Workers can update progress.
- Resolution evidence can be uploaded.
- Officers can verify resolution.
- Citizens can track complaints.
- Citizens can provide feedback.
- Overdue complaints can be identified.
- Reports can be generated.
- Role-based access works correctly.
- AI modules provide useful recommendations without preventing normal complaint processing.

21. Future Enhancements

- AI-powered image classification.
- AI-powered multilingual complaint processing.
- AI chatbot for citizens.
- Voice-based complaint registration.
- Advanced duplicate complaint detection.
- Predictive maintenance.
- Geographic complaint heat maps.
- Mobile application for Android/iOS.
- SMS and WhatsApp notifications.
- Integration with existing government/municipal systems.
- Advanced analytics and predictive dashboards.
- Automated municipal resource allocation.

22. Conclusion

The Municipal Complaint Management System provides a centralized solution for reporting, managing, assigning, tracking and resolving municipal complaints. The system connects citizens with municipal authorities and field workers through a structured complaint lifecycle. It improves transparency through complaint tracking, resolution evidence, feedback and status history, while dashboards and reports help authorities monitor performance and identify problem areas.

AI-assisted classification, priority prediction and duplicate detection can further improve efficiency while keeping final decisions under human supervision. The proposed system is technically, economically and operationally feasible and can serve as a foundation for a scalable smart-city civic service platform.